

to be one of the newest frontiers of innovation and development. Major types of such wearables include fitness trackers, smartwatches, ECG monitors, blood pressure monitors, biosensors and sugar-level monitors, ear-worn devices, head-mounted VR devices, and smart clothing. The wearable devices' market is differentiated by product type, end-use industry (industrial, media and entertainment), and connectivity medium.

Growth of the wearable market

Technology advancements, decreasing price of manufacture, augmented health awareness, small size and ease of use, and large brands associating with the wearables are some of the pivotal factors pushing the growth of wearable technology. A lot of industries such as healthcare, defence, entertainment, and others have adopted this wearable technology. But consumers are generally familiar with only activity trackers and smartwatches, which are supposed to provide them health benefits.

As obesity has become a major issue in many regions, people are buying wearable devices for healthcare and fitness-related data. However, these devices are also popular with teenagers and those in their twenties, who don them more for style statement than healthcare.

Internationally, style, fashion, and convenience are some of the factors and user preferences that are contributing to the growth of wearables. Most OEMs (original equipment manufacturers) have understood these factors and are therefore introducing

Experts' views

The smart wearable market was valued at around 200 million units in 2019 and is expected to double by 2022. The wearables segment aims to influence the fields of health and medicine, fitness, aging, disability, education, transportation, enterprise, finance, gaming, music, etc. Keeping in mind our key vision is to think smart and work smart, we challenge the norm by creating high-quality products that are innovative and serve the purpose of enhancing the way people live.

—**Manoj Kumar Pansari, chairman and managing director, Astrum**

Apart from hearables, fitness devices including bands and watches have been a strong factor in the growth of wearables in India. Though, India's wearables market is largely dominated by the basic devices. But this is helping in setting the base for the adoption of fitness devices.

As the global wearables market has shifted to smartwatches, India is still moving largely on bands. However, a clear shift is now seen that people are proactively using these devices for weight control and physical activity monitoring. Also, they have started asking for devices that can help them in monitoring some of the lifestyle diseases with better accuracy, forcing vendors to launch sophisticated devices in India. Also, they want devices which are good in look and feel and can go with their lifestyle.

In India, so far, the adoption has been limited to people who are tech-friendly and young. We haven't seen the adoption of child watches and the usage of smartwatches from elderly people much in India. Though, slowly, we see people have started gifting these devices to their parents, but that is a very small percentage of the overall market.

—**Jaipal Singh, associate research manager, Client Devices, IDC India**

Of course, wearables need to be sophisticated from two perspectives. One is this is technology being worn on the body, so has to be of the best quality to offer the best of safety to users wearing it. It should not result in any health hazard or accident. Second, wearables should give original experience. For instance, in the case of buds, it should give the best music/audio experience. In some cases, wearables need to have advanced software and computing capabilities to drive AI and other advanced technology use cases.

Right now from the Indian perspective, I would say the biggest challenge is that it's very fragmented with too many players, many of whom are not credible enough, and there can be no assurance that they are supplying good quality and value for ownership among consumers. Youth will continue to drive the sales, though these are also popular amongst professionals of all ages, including the sportspeople.

—**Faisal Kawoosa, founder, techArc**

numerous devices from time to time in various price segments, based on requirements in the market. The Internet of medical things (iOMT) is an emerging segment, which is expected to grow at a significantly high rate in next few years. Market research indicates that the global market of medical wearable technology will be worth \$8 billion this year.

According to market research agency IDC India, the Indian market for wearable devices grew 30.9 per cent in 2019. Apple, Samsung, and Fossil emerged as the most popular wearable brands in India last year. The industry is on track for a 25.8 per cent yearly increase in sales volume, having produced about 255 million units in

2019, according to an estimate.

The global market of wearables is witnessing 82.3 per cent growth, and it reached a new high of 118.9 million devices in the fourth quarter of 2019 according to another estimate. In the coming years, it is the healthcare sector that is expected to drive the entire wearable devices and technology market, which is also because there is a huge availability of cutting-edge products for health and fitness care.

Challenges

The wearables technology is constantly evolving, with new products and features faster than ever. But this technology is still in transition stage and improving every day. There are lot of expectations from wearables.

The biggest challenge facing wearables is to include new health-monitoring features in a small form factor that one can wear on wrist. Apart from the wristwear, other form factors haven't grown much.

But the market leaders of smart-

Gartner Forecast for Shipments of Wearable Devices Worldwide

Device (in million)	2017	2018	2019	2022*
Smartwatch	41.50	53.00	74.09	115.20
Head-mounted display	19.08	28.40	34.83	80.18
Smart clothing	4.12	5.65	6.94	19.91
Ear-worn device	21.49	33.44	46.12	158.43
Wristband	36.00	38.97	41.86	51.73
Sports watch	18.63	19.46	21.28	27.74
Total	140.82	178.91	225.12	453.19

*Estimated